Problem Pack: Square Roots and Square Applications from MATHEMATICS 8 • UNIT 2 REVIEW

Question 6

Problem: 6. Complete each inequality with the two consecutive whole numbers that the square root lies between. a) < √40 < b) < √55 < c) < √110 <

Student Work:

a) 6^2 = 36 < 40 < 49 = 7^2 < √40 < 7
b) 7^2 = 49 < 55 < 64 = 8^2 < √55 < 8
c) 10^2 = 100 < 110 < 121 = 11^2 < √110 < 11

Feedback

Question 7

Problem: 7. Approximate each square root to the nearest tenth. Use nearby benchmarks or nearby tenths to justify your answer. a) √45 b) √83 c) -√72

Student Work:

a) 6.7^2 = 44.89, 6.8^2 = 46.24 √45 = 6.7
b) 9.1^2 = 82.81, 9.2^2 = 84.64 √83 = 9.1
c) 8.4^2 = 70.56, 8.5^2 = 72.25 -√72 = −8.5

Feedback

In question 7, part c went wrong when you picked the closer tenth because 72 is closer to 70.56 than 72.25. Remember that you want to check which squared decimal is closest to your target number. Try looking at the distance between 72 and your two squared options again to pick the right tenth.

Question 8

Problem: 8. A square garden has an area of 52^2. a) Write the exact side length. b) State whether the side length is rational or irrational. c) Approximate the side length to the nearest tenth. d) State the two consecutive integers it lies between.

Student Work:

a) A = s^2 = s^2 s = √52
b) irrational
c) √52 = 7.211
d) 7^2 = 49 < 52 < 64 = 8^2 < √52 < 8

Feedback

Question 9

Problem: 9. Find the area of a square with each side length. a) 5 b) 1.2 c) 3/4 d) √6

Student Work:

a) A = 5^2 = 25^2
b) A = 1.2^2 = 1.44^2
c) A = (3/4)^2 = 9/16^2
d) A = (√6)^2 = 6^2

Feedback

Question 10

Problem: 10. a) A square has a perimeter of 28. Find its side length and its area. b) A square has an area of 45^2. Find its exact side length and its exact perimeter.

Student Work:

a) 4s = 28 s = 7 A = s^2 = 7^2 = 49^2
b) A = s^2 = s^2 s = √45 = 3√5 P = 4s = 12√5

Feedback

Question 11

Problem: 11. The figure is made from two squares. Use each area to determine the side lengths, then find the perimeter of the outside edge only. a) Find both side lengths. b) Identify the shared side that is not counted on the outside. c) Find the perimeter of the entire shape.

Student Work:

Student Work:

Feedback

In question 11, the student work is missing completely. You need to find the side lengths from the areas and then add up the outer edges to get the perimeter. Try writing out your steps for finding the sides and outer perimeter to finish the problem.